Innovation Diffusion Analysis: Garmin Index Sleep Monitor

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Step 1: The Chosen innovation

Garmin Index Sleep Monitor, from TIME's Best Inventions of 2025. Link: https://time.com/collections/best-inventions-2025/7318552/garmin-index-sleep-monitor/

Step 2: Look-alike innovation

Look-alike: Fitness trackers (Fitbit era).

-- Functionality and Technology To analyze the Garmin Index Sleep Monitor, I am using early early-stage Fitbit wearable trackers (such as the original Fitbit Tracker and Fitbit One) as my historical benchmark. Functionally and technologically, both products share the same core value proposition: screenless, lightweight body sensors that capture passive biometric data and sync it wirelessly to a companion smartphone app for interpretation. The Garmin Index Sleep Monitor is worn on the upper arm to track sleep stages, heart rate variability, breathing variations, and skin temperature to generate a personalized sleep score and wake-up alarm. Similarly, early Fitbits recorded health metrics like daily steps, calories, and basic sleep patterns, transmitting the data to a central portal. Both devices rely on the seamless integration of non-intrusive hardware sensing and software analytics to give users actionable health insights.

-- Market Impact and Adoption Dynamics From a market perspective, early Fitbit devices offer a clear historical precedent for how consumer health trackers diffuse. Fitbit drove rapid adoption partly through community features and social sharing, which created a strong peer-influence effect—directly mirroring the imitation coefficient (q) in the Bass diffusion model. However, while early Fitbits established the mass market for wearable activity trackers, the Garmin Index Sleep Monitor targets a more specialized niche. Priced at $169.99, it specifically solves the discomfort of wearing wristbands to bed for people who already track their health metrics. Because standalone activity trackers were eventually absorbed into general-purpose smartwatches, the Garmin Index functions as a targeted add-on device rather than a mass-market category creator. Consequently, while the diffusion curve will likely share a similar imitation pattern (q), its overall market potential (M) will be significantly smaller than that of early fitness wearables.

Step 3: Historical data

Data source. The look-alike series is the number of Fitbit devices sold worldwide per year from 2010 to 2025, in thousands of units. It comes from Statista (2026), which compiles it from Fitbit and Business of Apps. The dataset has 16 annual observations and is stored in `data/statista_fitbit_units_sold.csv`.

Year	Units sold (thousands)	Year	Units sold (thousands)
2010	58	2018	13,939
2011	208	2019	15,988

Year	Units sold (thousands)	Year	Units sold (thousands)
2012	1,279	2020	10,800
2013	4,476	2021	10,600
2014	10,904	2022	9,200
2015	21,355	2023	6,600
2016	22,295	2024	5,500
2017	15,343	2025	4,300

Shape of the series. Sales grew from 58 thousand units in 2010 to a peak of 22.3 million in 2016, then declined to 4.3 million in 2025. In total, about 152.8 million devices were sold over the period. The rise to 2016 is the part of the series that follows the Bass curve; the decline afterwards is analysed in Step 4.

Why this series matches the look-alike. It covers the full life of a wearable tracker from launch to maturity and decline, with yearly unit sales, which is the input the Bass model needs. It is also worldwide, which matches the global scope chosen in Step 6.

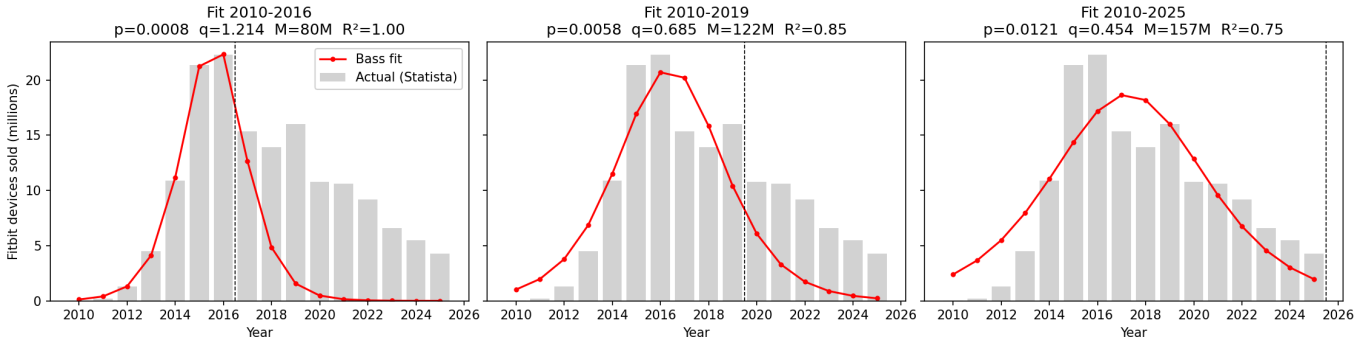
Limitations. The series counts devices sold, not distinct users, so repeat purchases and replacements are included. The values for 2020 to 2025 are rounded to the nearest 100 thousand units, while earlier years are given more precisely.

Step 4: Bass model parameters

Method. I fitted the discrete Bass diffusion model to the yearly Fitbit sales series (Statista, 2010–2025) using nonlinear least squares (scipy `curve_fit`). In the model, sales in year t are M × [F(t) − F(t−1)], where F(t) = (1 − e^(−(p+q)t)) / (1 + (q/p)·e^(−(p+q)t)), p is the coefficient of innovation, q is the coefficient of imitation, and M is the market potential (total units that will ever be sold). Code: `step4_estimate_bass.py`.

Choice of fitting window. Fitbit sales peaked in 2016 and then declined as smartwatches replaced stand-alone trackers. The Bass model describes a product that rises and fades on its own, so it cannot represent displacement by a competing technology. I therefore tested three windows.

Window	p	q	M (million units)	R^2
2010–2016	0.0008	1.214	80.4	0.9996
2010–2019	0.0058	0.686	122.2	0.853
2010–2025	0.0121	0.454	157.0	0.754



Interpretation. The 2010–2016 fit is almost perfect ($R^2 \approx 1$) but estimates three parameters from only seven observations, so the fit is nearly guaranteed and the parameters are implausible (p = 0.0008, q = 1.21). It also predicts sales falling to zero by 2019, which did not happen. The full-sample fit (2010–2025) has weaker fit (R^2 = 0.75) because the real decline after 2016 was slower than the model allows, its parameters (p = 0.012, q = 0.454) are close to the average values reported in a meta-analysis of 213 diffusion studies (p = 0.03, q = 0.38; Sultan, Farley and Lehmann, 1990). I use the 2010–2025 estimates as the main result and report the other two as robustness checks.

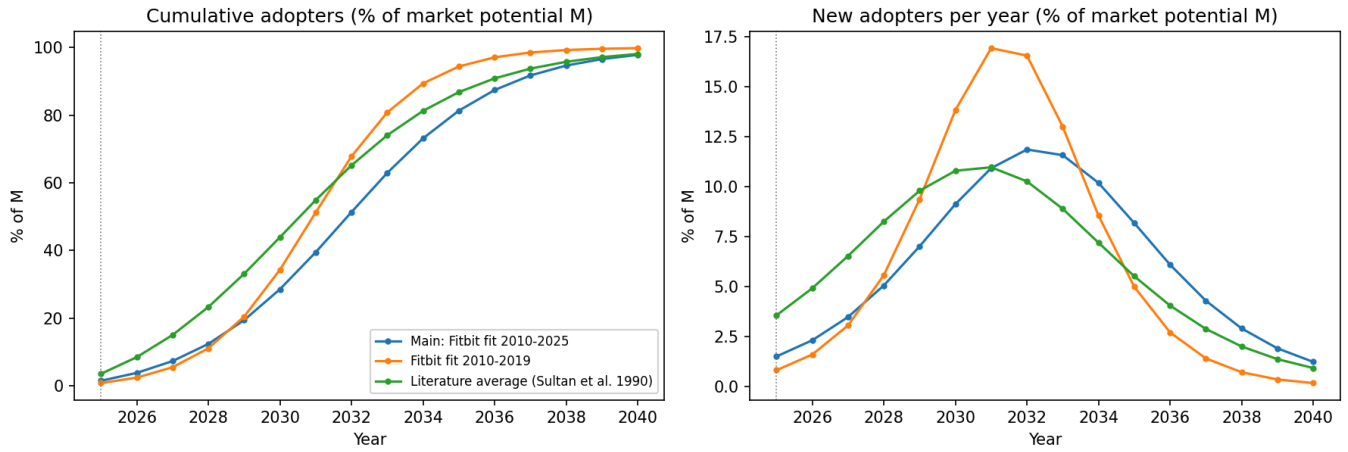
With p = 0.012 and q = 0.454, imitation is about 37 times stronger than independent adoption (q/p = 37.5). The model implies sales peak around 2017 and a market potential of about 157 million units.

Limitation. M counts units sold, not unique customers. Fitbit sales include repeat purchases and replacements, so M overstates the number of distinct adopters.

Step 5: Predicted diffusion

Method. The Garmin Index Sleep Monitor has no sales history, so I apply the Bass parameters estimated for the look-alike (Fitbit, 2010–2025: p = 0.0121, q = 0.4542) to Garmin. Time t = 1 is the launch year: Garmin launched the product on 18 June 2025 at a US price of $169.99 (Vice, 2025; Tom's Guide, 2025), so 2025 is year 1. Because M (market potential) is estimated separately in Step 7, the forecast here is expressed as a share of M. Code: `step5_forecast_garmin.py`.

Year	Cumulative adopters (% of M)	New adopters in year (% of M)
2025	1.5%	1.5%
2027	7.3%	3.5%
2030	28.6%	9.1%
2032	51.4%	11.9%
2035	81.3%	8.2%
2037	91.7%	4.3%
2040	97.8%	1.2%



Result. Under the main scenario, annual adoption peaks in 2032 at about 11.9% of M, half of the market is reached in 2032 and 90% by 2037.

Sensitivity. To test how much the forecast depends on one set of parameters, I repeated it with (a) the Fitbit fit on 2010–2019 data (p = 0.0058, q = 0.686) and (b) the average parameters from a meta-analysis of 213 diffusion studies (p = 0.03, q = 0.38; Sultan, Farley and Lehmann, 1990).

Scenario	Peak year	50% of M	90% of M
Main: Fitbit 2010–2025	2032	2032	2037
Fitbit 2010–2019	2031	2031	2035
Literature average	2031	2031	2036

The timing of the peak is stable across scenarios (2031–2032), which gives confidence in the shape of the curve.

Limitations. The look-alike assumption is the main one. Fitbit was a stand-alone wrist tracker, while the Index Sleep Monitor is a niche accessory that works best alongside a Garmin watch, so its adoption could be slower or tied to Garmin's existing customer base. The first period is also only a partial year (the launch was in June). The forecast gives the shape of diffusion; the number of adopters depends on M, estimated in Step 7.

Step 6: Scope

Decision. I analyse the diffusion of the Garmin Index Sleep Monitor **worldwide**. The choice rests on four points.

1. **The look-alike data is global.** The Fitbit series used to estimate the Bass parameters counts devices sold worldwide (Statista, 2026; Fitbit, 2016, 2020). The estimated p and q therefore describe global diffusion, and applying them to one country would mix two different scales.

2. **Garmin is a global business without a dominant country.** Garmin reported net sales of $7.25 billion in fiscal 2025, of which about 48% came from the Americas, 38% from EMEA and 14% from APAC (Garmin, 2026). The largest region is below half of sales, and "Americas" covers more than the United States, so a single-country analysis would cover well under half of Garmin's market.

3. **The product was launched in many markets.** The Index Sleep Monitor was priced at $169.99 in the United States and £149.99 in the United Kingdom (Tom's Guide, 2025), €169.99 in Europe (TechRadar,

2025) and ₱10,290 in the Philippines (TechPorn, 2025).

4. **Global data is available for the estimates in Step 7.** For example, IDC reports that 145.7 million wearable devices were shipped worldwide in the first quarter of 2026 (IDC, 2026). Consistent country-level data would be harder to source, and every number in this analysis needs a credible reference.

Limitations. However, a global scope hides differences between countries. Adoption speed and market conditions vary by region; for example, IDC reports that government subsidies pushed China's wrist-worn device shipments up 36.9% in 2025, far above the global growth rate (IDC, 2025). The global forecast should therefore be read as an average across markets. As an extension, the global number of adopters could be split by region using Garmin's regional revenue shares.

Step 7: Adopters by period

Method. Step 7 has two parts. First, I estimate the market potential M with a Fermi calculation. Second, I multiply M by the diffusion shape from Step 5 (p = 0.0121, q = 0.4542, t = 1 in 2025), so that adopters in year t equal M × [F(t) − F(t−1)]. One adopter means one band sold. Code: `step7_estimate_adopters.py`.

Fermi estimate of M. M = Garmin users × share who track sleep × share who buy a dedicated band.

Layer	Low	Base	High	Source
Active Garmin Connect users worldwide (millions)	42	45	49	the5krunner (2026)
Share of users who track sleep 21%	26%	48% Low: CCS Insight, cited in Raconteur ([YEAR]); Base: Rock Health (2025); High: AASM (2025)		
Share of those who buy a separate band	5%	10%	20%	Assumption of this report
M (million units)	**0.44**	**1.17**	**4.9**	Product of the layers

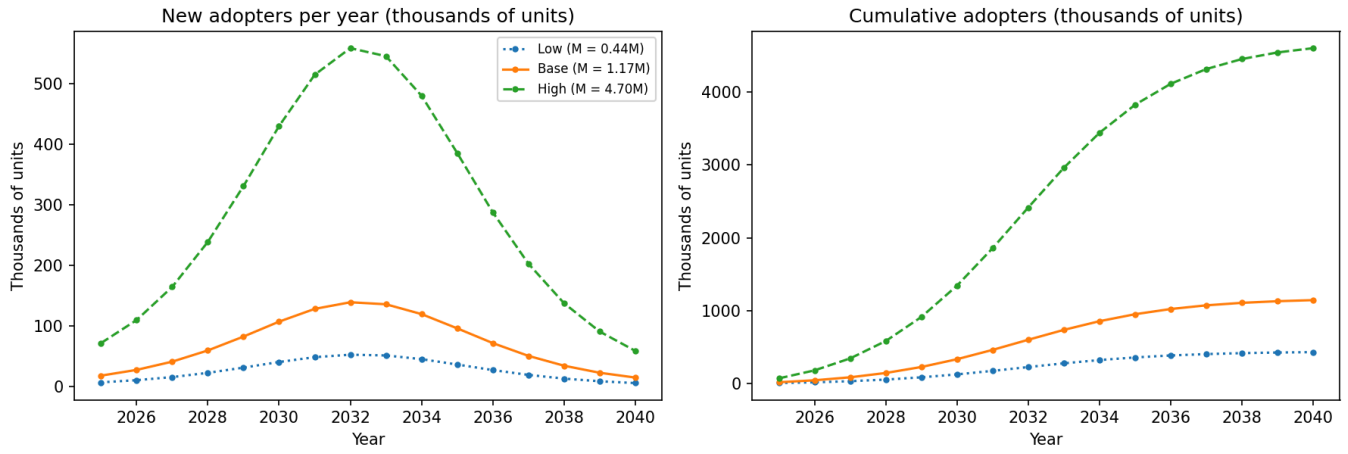
The target group is Garmin users, because Garmin markets the band to people who wear a Garmin device during the day (Best Buy, 2025). Non-Garmin buyers are excluded, so M is conservative. Garmin does not publish a Connect user count; the5krunner's estimate is the only available figure and is treated as an estimate. The share who buy a band has no source: buying a second $169.99 device is a much smaller group than users who track sleep, so I assume 10% and test 5% and 20%. The Rock Health share comes from a US survey of wearable owners. The AASM figure (48%) covers all US adults who use any sleep-tracking device, so the high case is an upper bound.

As a plausibility check, the base-case M of 1.17 million units equals about 5.7% of the 20.7 million units Garmin shipped in 2025 (Garmin, 2026), and the peak of about 139 thousand units a year is small next to the 4.9 million smart rings IDC forecasts for 2026 (IDC, 2026), consistent with a niche accessory.

Adopters by year (base case, thousands of units).

Year	New adopters	Cumulative adopters
2025	17.8	17.8
2027	40.8	85.8
2030	106.9	334.2
2032	138.8	600.9
2035	95.8	951.6
2037	50.4	1,073.4
2040	14.6	1,144.6

Scenario	M (million)	Peak-year new adopters (thousands, 2032)	Cumulative by 2040 (thousands)
Low	0.44	52	431
Base	1.17	139	1,145
High	4.70	558	4,602



Interpretation. In the base case about 600 thousand bands are sold by 2032, when annual sales peak, and about 1.14 million by 2040. The timing is fixed by the Bass parameters; the size depends on M, which varies about ten-fold between the low and high cases. The share who buy a band is the largest source of that spread.

Limitations. (1) The share who buy a band is an assumption. (2) The number of Garmin users is an external estimate. (3) The survey shares come from different samples, mostly in the United States, and are applied worldwide. (4) M is fixed, although Garmin's user base is growing, which would raise M over time. (5) The first period is a partial year (launch in June 2025). (6) M counts units sold, not distinct people.

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